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Department of Artificial Intelligence and Data Science Engineering

Question bank

Unit II Dictionaries and Query Processing

Sr No	Question	Blooms Level
1	Define the components of an inverted index and explain their roles.	BT2
2	Compare and contrast Sort-Based and Hash-Based dictionaries used in Information Retrieval.	BT4
3	Illustrate the process of constructing an index using the in-memory and sort-based method.	BT3
4	Explain the differences between Static and Dynamic Inverted Indexes. When would you prefer each?	BT 5
5	Describe the Index Life Cycle. Why is it important in the context of search engine efficiency?	BT 2
6	Analyze the advantages and limitations of Document-at-a-Time vs Term-at-a-Time Query Processing/	BT4
7	Design a small inverted index for a given set of three documents. Show the posting list and dictionary?	BT6
8	How does Impact Ordering improve query processing speed in ranked retrieval systems? Explain with an example?	BT2
9	Evaluate the effectiveness of merge-based and disk-based index construction in large-scale IR systems?	BT5
10	Propose a strategy to optimize query processing using pre-computed score contributions. Justify your approach.	BT6

Anagha Jawalkar/Rutuja Tikait/Chaitrali kardile

Subject Teacher

Dr.Mithra Venkatesan

HOD [AI&DS]